In[1]:= $$A = -130;\ B = 0.368;\ \lambda = A/B;\ EE = 600.$$

Out[1]= 600.

In[2]:= $$X[J\_] := \sqrt{4 * (J + 1/2)^2 + \lambda * (\lambda - 4)}$$

In[3]:= $$Y[J\_] := \sqrt{(J + 3/2) * (J - 1/2)}$$

In[4]:= $$\theta = \frac{A + 2 * B}{\sqrt{2}}$$

Out[4]= -91.4035

In[5]:= $$\xi[J\_] := \sqrt{2} * B * (J + 1/2)$$

In[6]:= $$\eta[J\_] := \sqrt{2 * (J + 3/2) * (J - 1/2)}$$

In[7]:= $$g0[J\_] := \frac{1}{J * (J + 1)} * \left(\frac{3}{2} + \frac{2 * Y[J]^2 - 1.5 * \lambda + 3}{X[J]}\right)$$

In[8]:= $$g0[3/2]$$

Out[8]= 0.804485

In[9]:= $$glp[J\_] := \frac{2 * Y[J] * (\lambda + 1)}{B * J * (J + 1) * X[J]^3} * \left((2 - \lambda) * \frac{(\theta + \xi[J]) * \eta[J]}{EE} - Y[J] * \frac{(\theta + \xi[J])^2 - \eta[J]^2}{EE}\right) + \frac{\sqrt{2}}{J * (J + 1) * X[J]} * \left(\left(-Y[J]^2 + \frac{X[J] - (2 - \lambda)}{2} * (J + 1/2)\right) * \frac{\theta + \xi[J]}{EE} + Y[J] * \left(-(J + 1/2) + \frac{X[J] + (2 - \lambda)}{2}\right) * \frac{\eta[J]}{EE}\right)$$

In[10]:= $$glp[3/2]$$

Out[10]= 0.00617717

In[11]:= $$glm[J\_] := \frac{2 * Y[J] * (\lambda + 1)}{B * J * (J + 1) * X[J]^3} * \left((2 - \lambda) * \frac{(\theta - \xi[J]) * \eta[J]}{EE} - Y[J] * \frac{(\theta - \xi[J])^2 - \eta[J]^2}{EE}\right) + \frac{\sqrt{2}}{J * (J + 1) * X[J]} * \left(\left(-Y[J]^2 + \frac{X[J] - (2 - \lambda)}{2} * (J + 1/2)\right) * \frac{\theta - \xi[J]}{EE} + Y[J] * \left((J + 1/2) + \frac{X[J] + (2 - \lambda)}{2}\right) * \frac{\eta[J]}{EE}\right)$$

In[12]:= $$glm[3/2]$$

Out[12]= 0.00629938

In[13]:= $$glp[9/2] - glm[9/2]$$

Out[13]= -0.000369752